Mini-project 2 solution

Here is a derivation of a cell-based model based on the 8 steps for model formulation covered at the start of class.

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# Steps in model formulation

1. Initial concepts and assumptions (boundary, state variables, sketches)
2. Balance/conservation equations
3. Constitutive equations (transfer or rate equations)
4. Linking equations
5. Spatial or other simplifications
6. Initial or boundary conditions
7. Governing equation
8. Analytical or numerical solution
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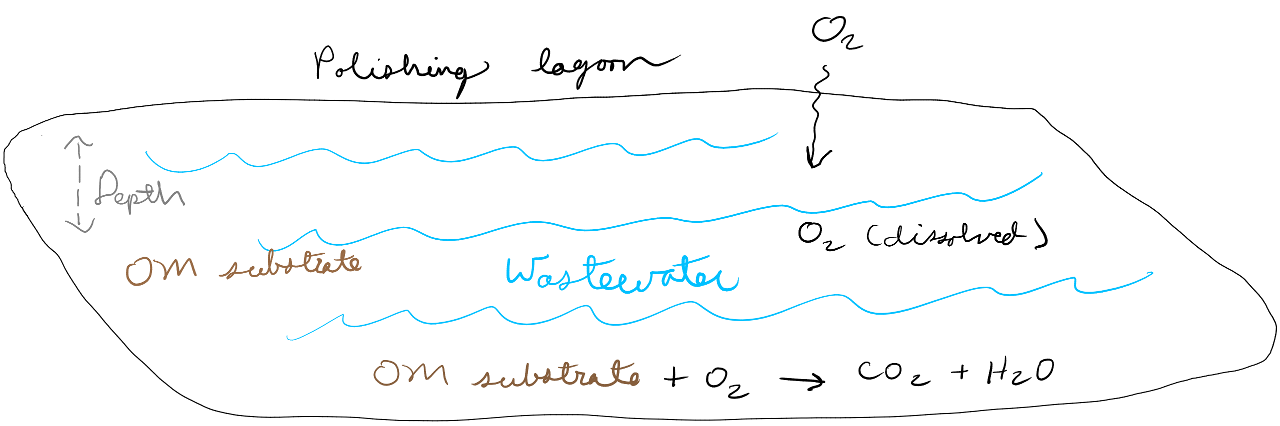
This is an alternative to the node-based approach.

Model formulation

1. Initial concepts and assumptions, including a sketch and the conceptual model

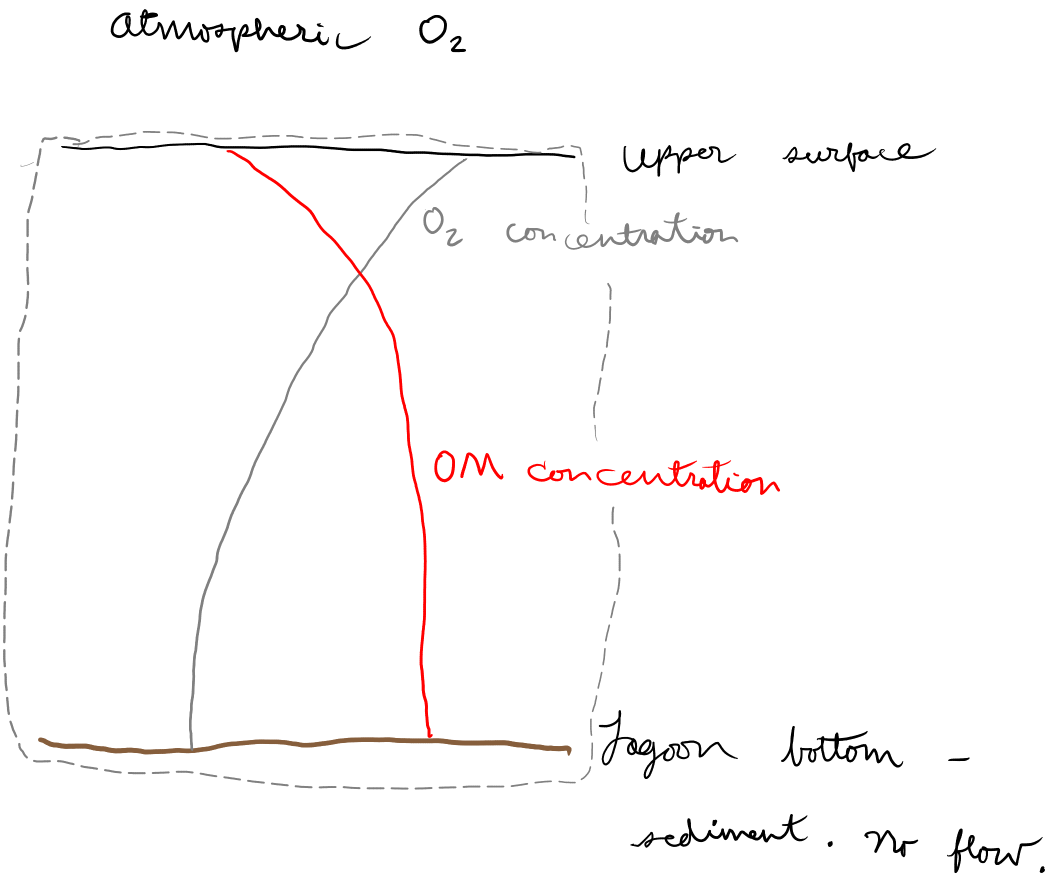
Sketch and conceptual model



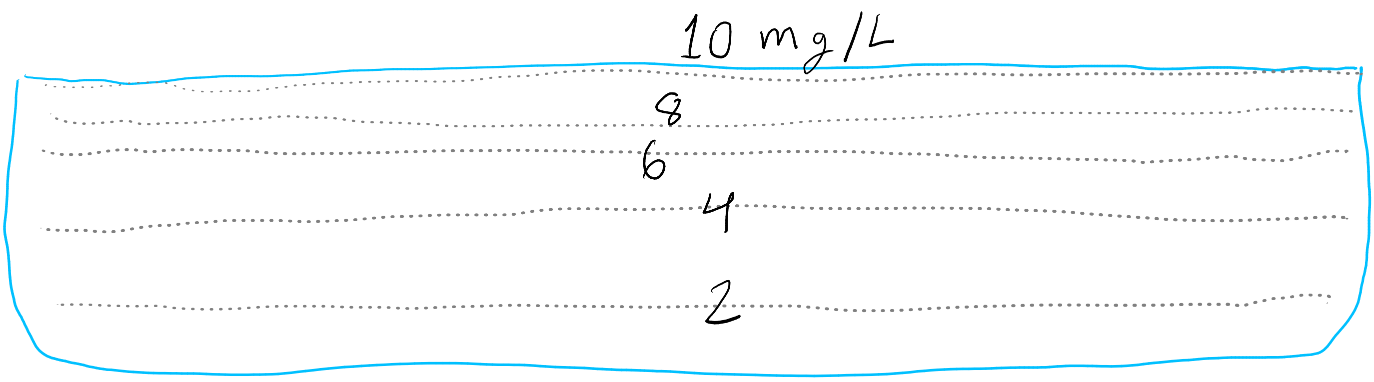
No inflow or outflow so run in batch mode.

State variables are concentrations of dissolved O2 and substrate OM.

We'll include some simplifications here.
Right from the start let's plan on developing a 1D model. This is a reasonable simplification.
Horizontal gradients in OM or O2 should not exist if the system is run in a batch mode
and well-mixed at the start. So our single dimension is depth.



Our model boundary is shown by the dashed gray line. We can simulate a column
of water without thinking about how wide it is (or its volume) because this is a 1D model.
So we assume that state variable values depend only on depth and time--they are always
uniform over horizontal space. So lines of constant O2 concentration might look
like this over the lagoon's width at some particular time, for example. We don't
have to think about horizontal space in this 1D model.



Because transport is thought to be by diffusion, and

* diffusion in liquid is much slower than in gas, and
* O_2 is not very soluble in water,

we can assume all mass transfer resistance is in the water. So we can use the saturation O_2 concentration at the top as a model boundary condition.

2. Balance/conservation equations

Here we are working with mass. We need a mass balance equation for each state variable.

Rate of O2 accumulation = O2 diffusion in - O2 consumption via reaction

Rate of OM accumulation = - rate of OM consumption via reaction

Let's use base units of kg, m, and s. So all the rates above are in kg/m3-s.

$kg\ m^{-3}\ s^{-1}$

But kg of what? Well, either O2 or OM. We are effectively using mass of oxygen or oxygen demand as our units.

that OM degradation is a second-order reaction. You can also assume that it takes 1 gram of oxygen to degrade 1 gram of OM. (In fact this is usually done in the wastewater treatment field because organic matter is expressed in "oxygen demand" units.)

That is convenient, because then we do not need
any stoichiometric coefficients and the rates of the two reactions are exactly the same.

3. Constitutive equations

We need equations for diffusion and reaction.

For diffusion, we have Fick's law. One possible approach here is to apply the steady-state form to each pair of model nodes or cells. An alternative is to jump right to the governing equation from the book--see step 7 below for that approach.

Fick's law, steady-state form

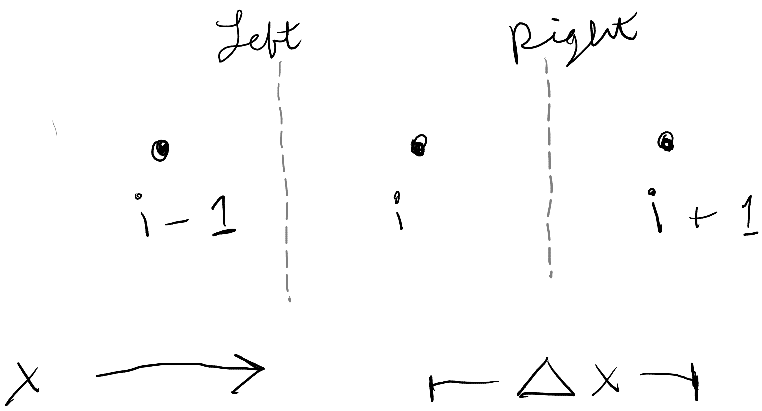
$$J = -D \cdot \frac{\Delta c}{\Delta x}$$

Units on flux J here are kg/m2-s.
c is concentration (kg/m3).

That applies to both of our solutes, O2 and OM.

In this 1D model, we can apply this equation to individual nodes or cells. This is shown below for a single node at position i, which could also be interpreted as the center (or average) of a single cell.

We'll separately think of flux to the central node from the left, and then from it to the right.



J_left = - D * (c[i] - c[i-1]) / dx

J_right = - D * (c[i+1] - c[i]) / dx

Those expressions hold for all nodes in the model. Probably we should have used "top" and "bottom" instead of left and right, but you get the idea.

For the reaction, we'll assume second-order, so we have:

$$r = k \cdot c_{O_2} \cdot c_{OM}$$

Here r = reaction rate in kg/m3-s, k is the second-order rate constant in m3/kg-s, and c are still concentrations (kg/m3)

4. Linking equations

You could argue that we don't need any linking equations, because our constitutive equations include our state variables directly. And we have a single reaction rate, because the stoichiometric coefficient is effectively unity (1).

But Fick's law just gives flux, while we need the time derivative of concentration. We could avoid thinking about all this by just using the governing equation from the book. But here we'll explicitly get equations for the diffusion contribution to the concentration derivative.

For this we have to think about cell volumes.

The rate of change in concentration due to diffusion is just the difference between diffusion in and out divided by the cell volume. Flux is per m2, so we really need to divide by area-normalized volume, which is dx.

dc/dt = (J_in - J_out) / dx

In case this use of dx for cell volume is not clear, let's instead do a more careful derivation, using some arbitrary cross-sectional area A. Now cell volume is A * dx. And we need to also multiply the net flux by the area to get mass flow rate, not flux, if we are dividing by the cell volume.

dc/dt = A (J_in - J_out) / (A dx)

And A cancels, giving the earlier equation.

Now, let's start with that equation and substitute in our equationa for Fick's law (orange below). We'll do this for one cell, but the resulting equation applies to all of them.

J_left = - D * (c[i] - c[i-1]) / dx

J_right = - D * (c[i+1] - c[i]) / dx

dc[i]/dt = (- D * (c[i] - c[i-1]) / dx
+ D * (c[i+1] - c[i]) / dx) / dx

Simplifying. . .

$$dc[i]/dt = D (c[i+1] - 2c[i] - c[i-1]) / dx^2$$

results in the central difference formula!

5. Simplifications. We've already made several important simplifications.
See step 1 and other parts.

6. Initial or boundary conditions

Initial conditions (that is, values of our state variables at the start) could vary. For the model validation part we do have some info:

3. Model validation

An engineer at the wastewater plant measured the concentration of dissolved oxygen near the bottom of a 10 cm deep lagoon over a couple months, starting right after filling. The measurement data are in the `meas_02.csv` file. The filling process effectively aerated the wastewater at the start, but as you can see in the measurements, dissolved oxygen concentration declined over the following days. The initial OM concentration was 0.02 kg/m3 (20 mg/L). Carry out graphical and quantitative model validation using these measurements.

So the initial O2 concentration should be at or near saturation. And initial OM 0.02 kg/m3.

For boundary conditions, we have no flux at the bottom. This is a Neumann type boundary condition (the value of the derivative is fixed, here at zero). We can define that mathematically like this (assuming that the bottom is the last index): dcdx(t, c = L) = 0

For the top, by assuming resistance is only in the liquid phase, we must have the saturation concentration of O2 as the upper boundary condition. This is a Dirichlet boundary condition. We can define this either directly at the boundary or 1 dx outside the boundary.
c(t, c = 0) = 10, or c(t, c = c_outside) = 10.

- The concentration of O_2 in pure water in equilibrium with the atmosphere is around 10 mg/L. You can use this value without explicitly simulating inter-phase mass transfer if you assume all the resistance to wastewater aeration in this system is present in the water, which is reasonable.

So we'll use 10 mg/L, which is 0.01 kg/m3.

But the upper boundary condition is different for OM; it is just like the bottom--no flux or Neumann with dcdx(t, c = 0) = 0.

7. Governing equations

Substitute the constitutive equations into the mass balance equations.

Rate of O2 accumulation = O2 diffusion in - O2 consumption via reaction

Rate of OM accumulation = - rate of OM consumption via reaction

We can write these for individual nodes or cells.

$$\frac{dc_{O_2}}{dt} = \text{rate of } O_2 \text{ accumulation}$$

$$dc_{O_2}[i] = \overbrace{D \cdot (c_{O_2}[i+1] - 2c_{O_2}[i] - c[i-1]) / \Delta x^2}^{\text{Diffusion part}} - \underbrace{k \cdot c_{O_2} \cdot c_{OM}}_{\text{Reaction part}}$$

And for OM

$$dc_{OM}[i] = \overbrace{D \cdot (c_{OM}[i+1] - 2c_{OM}[i] - c[i-1]) / \Delta x^2}^{\text{Diffusion part}} - \underbrace{k \cdot c_{O_2} \cdot c_{OM}}_{\text{Reaction part}}$$

Now, we could have skipped some steps by starting with this governing equation from the course text book:

A.1 Rectangular coordinates

Mass balance for component A (ρ and D_{AB} constant)

$$\frac{\partial C_A}{\partial t} + v_x \frac{\partial C_A}{\partial x} + v_y \frac{\partial C_A}{\partial y} + v_z \frac{\partial C_A}{\partial z} = D_{AB}\left(\frac{\partial^2 C_A}{\partial x^2} + \frac{\partial^2 C_A}{\partial y^2} + \frac{\partial^2 C_A}{\partial z^2}\right) + R_A.$$

For a 1D model we can eliminate all y and z terms. . .

A.1 Rectangular coordinates

Mass balance for component A (ρ and D_{AB} constant)

$$\frac{\partial C_A}{\partial t} + v_x \cancelto{0}{\frac{\partial C_A}{\partial x}} + v_y \cancelto{0}{\frac{\partial C_A}{\partial y}} + v_z \cancelto{0}{\frac{\partial C_A}{\partial z}} = D_{AB}\left(\frac{\partial^2 C_A}{\partial x^2} + \cancelto{0}{\frac{\partial^2 C_A}{\partial y^2}} + \cancelto{0}{\frac{\partial^2 C_A}{\partial z^2}}\right) + R_A.$$

$$\frac{\partial C_{O_2}}{\partial t} = D \cdot \frac{\partial^2 C_{O_2}}{\partial x^2} + r$$

Then discretize and use the central difference formula.

dcdt = D * (c[i+1] - 2 * c[i] + c[i-1])/dx^2. At the boundaries we will have special cases where we substitute a ghost point (for neumann BC). This is shown in the code.

8. The solution

Even with the all the information given above, actually programming the model into a Python function is tricky. We have to think about how to actually implement the boundary conditions and the syntax for indexing, among other challenges.

See the modules for example implementations.